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ORIGINAL ARTICLE

Validating large language model—assisted data extraction from clinical notes

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Background: Health care professionals face increasing documentation burdens, which can compromise efficiency and patient safety. Large language models (LLMs) may offer a scalable solution by automating data extraction from unstructured clinical notes. This study evaluates the accuracy and clinical impact of structured data extraction by an LLM compared with manual extraction by physicians in the context of head and neck oncology consultations.

Patients and methods: This was a prospective validation study comparing LLM-powered data extraction with manual extraction. Clinical documentation from 60 patients (1482 pages) was analyzed. Data were extracted for 29 clinically relevant categories by two physicians and a pretrained open-source LLM. Six clinical experts evaluated 2555 extracted values in a two-step procedure: first, a blinded binary Match/Non-Match assessment between LLM output and human consensus; and second, categorization of all Non-Matches into predefined error types (*Incorrect*, *Incomplete*, *Missing*, *Hallucination*, or *Overcomplete*). These expert-assigned labels formed the basis for accuracy, precision, recall, and F1 scores. Extraction times were compared using a paired Student's *t*-test, and error impact was scored on a 5-point Likert scale.

Results: LLM-powered extraction achieved accuracies between 74% [pathology: 95% confidence interval (CI) 65% to 82%] and 90% (patient characteristics: 95% CI 87% to 94%). Manual extraction showed 29% interobserver disagreement (95% CI 25.19% to 32.74%). Of 2555 extracted values, 68 were rated as high-impact errors, although evaluator assessments varied widely. Hallucinations were rare (0.16%) and low impact. LLM extraction reduced average time per case from 8.6 minutes to 1.9 minutes ($P < 0.001$).

Conclusion: LLMs can support clinical workflows by reducing documentation time and maintaining acceptable accuracy, provided that human oversight is ensured. These findings support further exploration of AI-assisted documentation tools in clinical practice.

Key words: administrative burden, information extraction, large language model (LLM), oncology, real-world data, validation

INTRODUCTION

High documentation load places a significant burden on Dutch health care professionals,[1,2] caused by numerous administrative tasks and limited time availability.[3] The introduction of electronic health records (EHRs) has further increased this burden, resulting in more time spent on administrative tasks and less on direct patient care.[4] This documentation burden is correlated with an increase in medical errors, threats to patient safety, lower documentation quality, high employee turnover, and health care professional burnout.[3,5,6]

Expectations for artificial intelligence (AI) in clinical settings are high, particularly regarding the potential to reduce administrative burden and streamline clinical workflows. These expectations are fueled by recent advancements in natural language processing (NLP) and the introduction of large language models (LLMs).[7,8] To date, however, most AI-related publications have focused on methodological development[9] or applications in preclinical research.[10-12] Before AI

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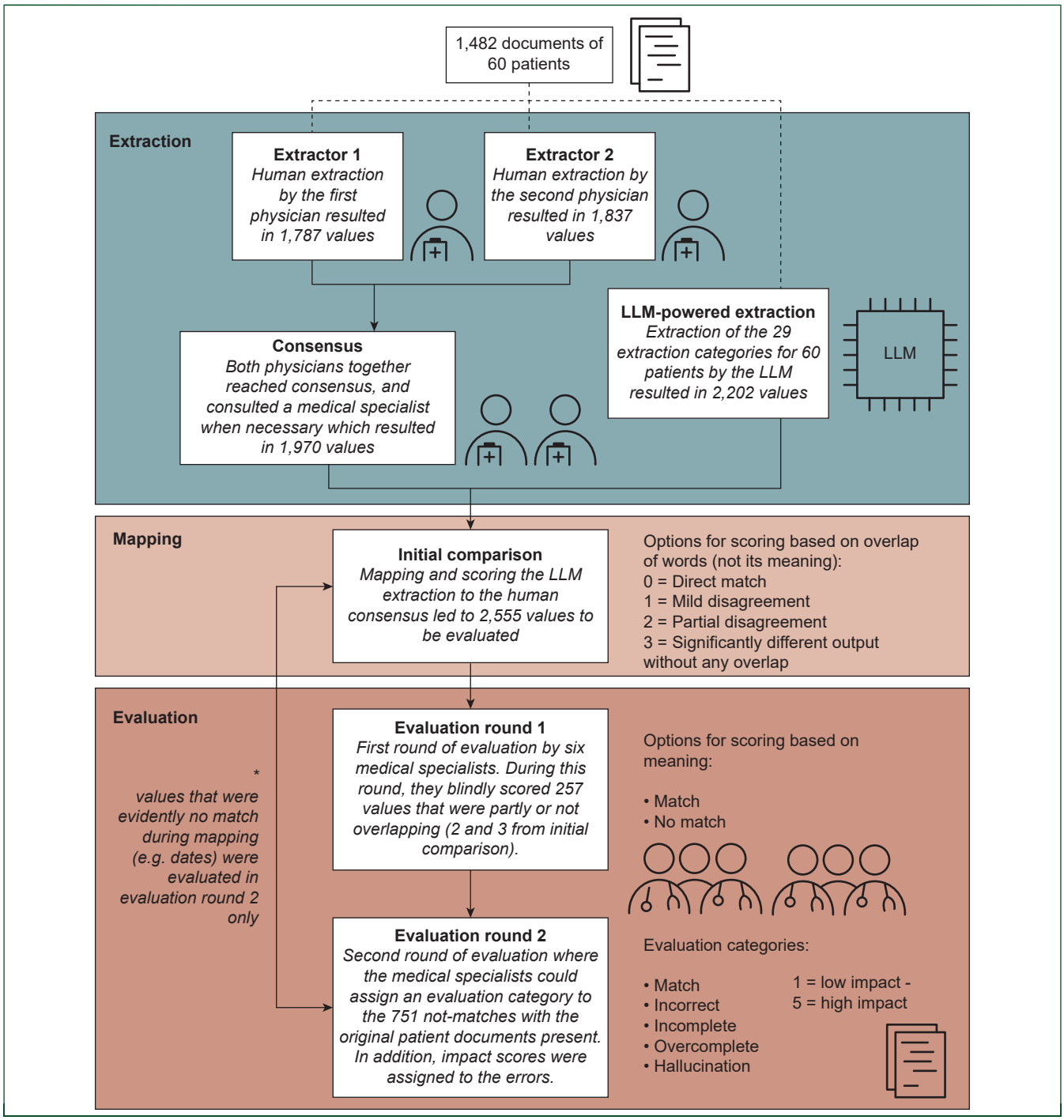


Figure 1. Validation workflow, including both human and LLM-powered extraction. The number of extracted and evaluated values is given per step. The number of extracted values differs between humans and the LLM, even though they were all given the same input documents, because they could format their responses differently (e.g. splitting or combining two conditions, or alternatively, by simply overlooking information). We decided not to penalize the formatting in this study. The blind evaluation step is referred to as evaluation round 1 and the in-context evaluation as evaluation round 2. On the right, the options for scoring per step are given. LLM, large language model. An example of the evaluation for one extracted value is shown in Supplemental Figure S1 (available at https://doi.org/10.1016/j.esmorw.2026.100718.).

can be meaningfully integrated into clinical practice, it is essential to validate tools for their clinical performance and usability, including safety and overall clinical effectiveness.

In oncology, clinicians rely on a wide range of information, including radiological findings, pathological characteristics, and patient history, before the first consultations with patients. These variables are essential for forming an initial clinical impression, determining diagnostic completeness, and planning subsequent diagnostic or therapeutic steps. Few studies have evaluated LLM-based extraction in real-world clinical workflows, especially for administrative purposes and in comparison with human performance.[13]

This study assessed the accuracy and clinical impact of LLM-powered data extraction in the Head and Neck Cancer

(HNC) department of Antoni van Leeuwenhoek, with the aim of supporting preparation for first consultations. We compared extraction results from clinical notes by a pretrained, open-source LLM to human consensus across 29 clinically relevant variables spanning radiology, pathology, and patient history. By evaluating both extraction performance and the clinical impact of any errors, this study provides an important first step toward responsible implementation of AI-driven documentation support in real-world oncology workflows.

PATIENTS AND METHODS

Study design

We assessed the accuracy of LLM-powered extraction from unstructured clinical notes by performing a prospective validation study using existing patient data. Figure 1 outlines the study workflow, which included manual extraction by two physicians as a reference and two extensive evaluation rounds by six medical domain experts. Full details of extraction counts, mapping steps, scoring options, and evaluation categories are shown in Figure 1 and explained in more detail in the next sections.

Study population

We selected 60 patients referred for an initial consultation at the department of Head and Neck Surgery and Oncology of Antoni van Leeuwenhoek, a specialized comprehensive cancer center, in the period of June 2023-June 2024. Patients were either newly referred cancer patients or patients receiving a second opinion at the hospital. We included consecutive patients who fulfilled the inclusion and exclusion criteria, up to a total of 60 patients. Patients were eligible for inclusion if at least one referral letter was available. Exclusion criteria were the availability of >10 documents per patient or patients with documents that were nearly empty or had poor scanning quality. Details regarding the study population and supporting documentation are outlined in Supplementary Table S1, available at https://doi.org/10.1016/j.esmorw.2026.100718.

Six patients were excluded due to the above criteria. Four patients were excluded due to a lack of referral documents. The remaining two were excluded due to the poor quality of the scanned documents. This low quality was caused by formatting issues, such as screenshots and large handwritten texts, which hindered optical character recognition processing, and subsequently, the generation of input for the LLM.

Manual data extraction workflow

To determine the human baseline standard, two medical physicians (co-authors AB and SD), experienced in conducting first consultations in HNC, manually extracted values in text fields for 29 categories from the documents of 60 patients. These 29 categories covered a wide range of information used in the preparation of the first consultation, including reason for referral, conducted medical tests, and general patient information such as allergies. Categories were identified in collaboration with physicians and nurse practitioners to ensure clinical relevance and adequate coverage of the information needed for first consultations. A complete overview of the categories, their descriptions, and formatting is provided in Supplementary Table S2, available at https://doi.org/10.1016/j.esmorw.2026.100718. Throughout the manuscript, we refer to the extracted information within each category as values. All data resulting from manual extraction were registered in Castor (version 2024.3), a secure clinical data management platform. For each extraction session, extraction times were recorded, which were cross-checked against Castor's metadata to ensure accurate measurement of the time spent on manual data entry.

After the manual extraction, the two physician-generated datasets were directly compared, and each pair of corresponding values was assigned a mapping score by the first author. Score 0 represented a direct match, 1 a mild disagreement (e.g. a small formatting difference), 2 a partial disagreement (e.g. incomplete information), and 3 significantly different output without any overlap. For the values with scores 2 and 3, consensus was reached during in-person meetings with the two physicians and a researcher. An additional expert was consulted when disagreements between the physicians' extractions could not be resolved.

LLM-powered extraction

Full details on the pipeline are described in the Supplementary Methods, available at https://doi.org/10.1016/j.esmorw.2026.100718.

Evaluation

The LLM-extracted output was compared to the human consensus across two rounds by six domain experts in Head and Neck Surgery and Oncology, referred to as evaluators. To enable comparisons, the mapping score (0-3, ranging from Direct Match to Significantly Different Output without Overlap) was assigned to the combinations of model output versus human consensus, as was done in the earlier human—human comparisons.

Because the model generates free-text outputs rather than discrete categorical predictions, extraction performance was determined entirely by expert-assigned labels obtained using a two-step evaluation procedure:

Step 1: binary assessment (Match/Non-Match). In the first round, each evaluator was assigned a distinct subset of 10 patients and asked to assign a binary score to the values mapped with a score of 2 or 3 between the LLM and human extractors: *Match* or *Non-Match*. The evaluation was blinded, meaning that evaluators were not told which response belonged to the model versus humans. The evaluators were given patients in a random order and had no access to the original patient documents. When scoring, the evaluators were instructed to consider only content, not format.

Step 2: ordinal error categorization for all Non-Matches. During the second evaluation round, each evaluator was

randomly assigned a distinct, nonoverlapping subset of 10 patients from the same cohort of 60, ensuring that these 10 patients were new to that evaluator compared with round 1. Each Non-Match was then assigned a predefined error category: *Incorrect*, *Incomplete*, *Missing*, *Hallucination*, or *Overcomplete* (Supplemental Table S3, available at https://doi.org/10.1016/j.esmorw.2026.100718). If for a certain category, both consensus and LLM output were absent, the category *Not mentioned* was assigned automatically. Together with *Match*, these seven mutually exclusive, expert-assigned labels constituted the multiclass input for quantitative analysis. The evaluation was unblinded for this round, and the original patient documents were provided. In addition to assigning the error categories, the evaluators rated the potential clinical impact of the error regarding the preparation for the initial consultation using a Likert scale ranging from 1 to 5. To further calibrate clinical impact, we compared LLM extraction performance to human variability.

The two evaluation rounds together yielded a single expert-assigned outcome label for each extracted value. These labels replaced the earlier mapping score (0-3) and formed the final seven-class outcome space used for quantitative performance analysis.

Statistical methods

The sample size for this study was chosen to provide a representative coverage of the entire study population to ensure reliable and generalizable results. We evaluated model performance using accuracy, precision, recall, and F1 score, with the human consensus serving as the reference standard. As the LLM does not produce categorical predictions, all performance metrics reported in this study should therefore be interpreted as expert-assigned evaluation metrics derived directly from these labels.

Accuracy, precision, recall, and F1 score were calculated within each extraction category by mapping the expert-assigned labels onto clinically meaningful true-positive, false-positive, false-negative, and true-negative groups: *Match* and *Overcomplete* were treated as true positives (correct information present, with or without additional content); *Incorrect* and *Hallucination* as false positives (incorrect or fabricated information); *Incomplete* and *Missing* as false negatives (insufficient or absent information); and *Not mentioned* as true negative. This multiclass taxonomy was feasible given the dataset size (2555 evaluated values), providing sufficient observations per error type for stable estimates. To quantify uncertainty in performance estimates, 95% confidence intervals (CIs) for accuracy and other performance metrics were obtained using nonparametric bootstrap resampling (100 000 samples).

As a post hoc exploratory analysis, we examined interobserver variability by comparing the two independent human extractions before consensus formation. Disagreement rates and corresponding 95% CIs were computed using the normal approximation: estimate $\pm$ 1.96 $\times$ standard error.

Comparisons of mean extraction time were conducted using paired two-tailed Student's *t*-tests. $P < 0.05$ was considered statistically significant.

Impact scores were first normalized to the evaluator-specific mean and subsequently averaged within each extraction category.

Ethics approval

This study did not fall under the scope of the Medical Research Involving Human Subjects Act, which was confirmed by the Antoni van Leeuwenhoek-Netherlands Cancer Institute Institutional Review Board (IRBd23-203). Patients gave consent for the secondary use of their data. The research was conducted in accordance with the Netherlands Code of Conduct for Research Integrity.

RESULTS

Overall performance and category breakdown

Overall performance across all categories was as follows: overall accuracy = 0.81; precision = 0.82; recall = 0.91; F1 = 0.86 (Table 1). All performance metrics reported here are expert-assigned evaluation metrics derived from the multiclass labeling framework described in the Statistical Methods. Performance varied substantially between categories (Table 1 and Figure 2). Categories related to *referral* achieved the highest F1 score (0.94, 95% CI 0.91-0.96), reflecting the relatively explicit and consistently formatted information in these documents. In contrast, categories such as *pathology* and *patient history* had the lowest F1 scores (0.81, 95% CI 0.72-0.88 and 0.81, 95% CI 0.72-0.87, respectively). After qualitative error analysis, we observed that these lower F1 scores in these categories are due to greater content variability, narrative complexity, and the presence of implicit details in these sections. Although F1 score differences were the most pronounced, similar trends were observed in precision, recall, and accuracy, indicating that category-specific characteristics influenced extraction performance.

Significant interobserver variability in manual extraction

We quantified interobserver variability by comparing the two human expert extractions before reaching a consensus and assigning mapping scores. Mapping scores of 2 and 3 were considered to represent a significant difference. For each patient (n = 60), we first computed the fraction of extracted values falling into the disagreement categories across all 29 extraction domains. Averaging these patient-level fractions yielded an overall human—human disagreement rate of 29.0% (95% CI 25.19% to 32.74%). Applying the same approach to the comparison between the human consensus and the LLM resulted in a disagreement rate of 41.4% (95% CI 38.33% to 44.46%). The higher disagreement rate between the LLM and the human extractors (41.4% versus 29.0%) indicates that the LLM diverged more from the consensus than human extractors did from each other. Nonetheless, the substantial proportion of values with partial or complete disagreement in both

Table 1. Expert-assigned accuracy, precision, recall, and F1 scores for the extracted values

Extraction category	Accuracy (95% CI)	Precision (95% CI)	Recall (95% CI)	F1 (95% CI)
Referral	0.89 (0.83-0.93)	0.91 (0.86-0.96)	0.97 (0.95-0.99)	0.94 (0.91-0.96)
Pathology	0.74 (0.65-0.82)	0.72 (0.61-0.83)	0.96 (0.91-0.99)	0.81 (0.72-0.88)
Radiology	0.75 (0.66-0.83)	0.76 (0.65-0.85)	0.91 (0.85-0.96)	0.82 (0.74-0.89)
Clinical chemistry laboratory	0.87 (0.67-0.98)	0.86 (0.59-1.00)	0.92 (0.76-1.00)	0.89 (0.70-0.98)
Patient history	0.76 (0.69-0.83)	0.85 (0.75-0.92)	0.78 (0.68-0.87)	0.81 (0.72-0.87)
Patient characteristics	0.90 (0.87-0.94)	0.81 (0.71-0.89)	0.92 (0.87-0.97)	0.86 (0.78-0.91)

Performance metrics were derived from the multiclass structure used in expert evaluation by multiple independent evaluators, consisting of *match* and the ordinal error categories (*Incorrect*, *Incomplete*, *Missing*, *Hallucination*, and *Overcomplete*). As detailed in the Statistical Methods section, experts first assigned a binary Match/Non-Match judgment and subsequently categorized all Non-Matches using this predefined taxonomy. Accuracy and other performance metrics were first calculated within each extraction subcategory, after which overall averages and bootstrap CIs were derived. The 95% CIs are estimated using nonparametric bootstrapping at the document level (resampling entire letters).
CI, confidence interval.

comparisons (Figure 3) highlights the inherent variability of the data extraction task, even among experts.

Many initial disagreements were resolved on closer inspection

Further review during evaluation rounds 1 and 2 showed that more than half of the LLM outputs initially scored as discrepant or partially overlapping were ultimately reclassified as valid matches (Supplementary Figure S2, available at https://doi.org/10.1016/j.esmorw.2026.100718). As we did not proceed with the evaluation of the extractors' output, we cannot draw conclusions about the error distribution of the human extractions.

Impact was not associated with performance

As well as scoring agreement between the LLM and human consensus, on the second pass of the evaluation, we asked evaluators to rate the clinical impact of any errors on a scale from 1 to 5. At this stage, the values were unblinded (evaluators knew which values originated from the human consensus and which from the LLM), and we observed considerable differences across evaluators (Supplementary Figure S3, available at https://doi.org/10.1016/j.esmorw.2026.100718.). In particular, we observed a notable deviation in scoring by evaluator 3, who was responsible for classifying errors as *high-impact* 61 of 68 times, while the remaining seven *high-impact* errors were generated uniformly among the rest of the evaluators. To account for this bias, we normalized impact scores; the resulting values are shown in Figure 4. The average impact of an error was 2.33.

High-impact errors did not correlate with low-performing categories. We also found that all evaluation categories (and therefore types of model errors) were classified as *high-impact* at least once. The defined error taxonomy and performance metrics alone are therefore insufficient to gauge the potential clinical consequences of extraction errors.

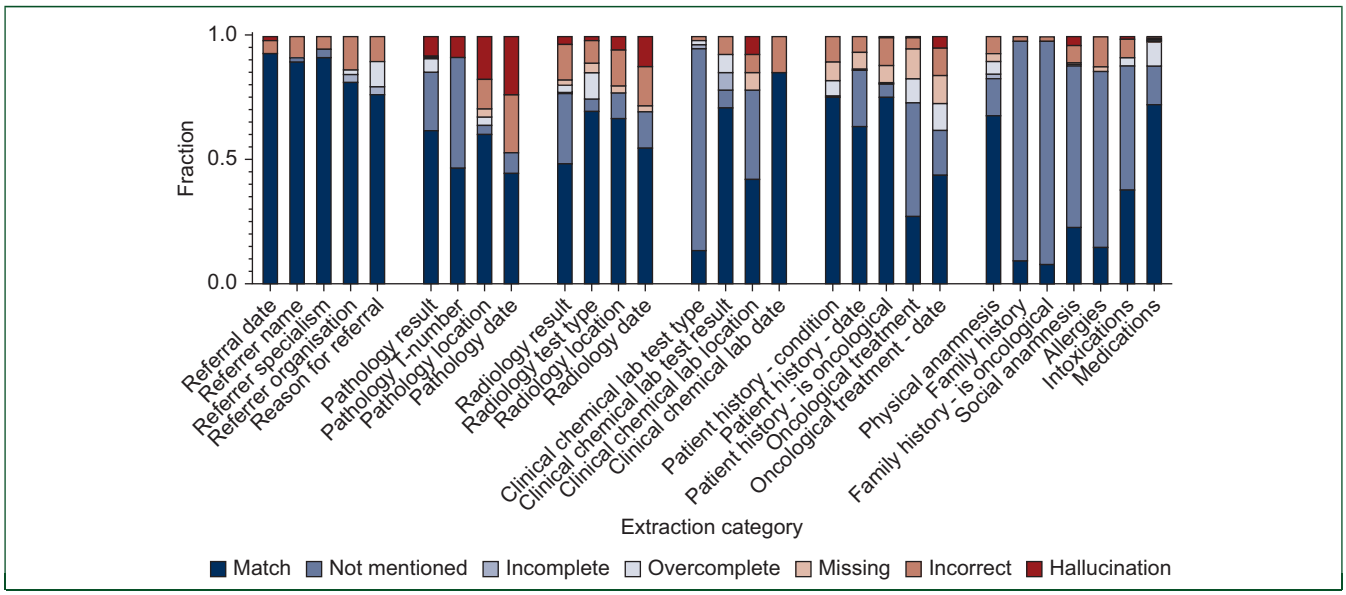


Figure 2. Distribution of error types for all categories after two rounds of expert evaluation. Large language model output was evaluated against consensus with the presence of the original documents. This means that the evaluators were allowed to go beyond the context of consensus to look for additional information. The expert-assigned error types were mapped to the confusion matrix, as noted in the Statistical Methods section. After these mappings, the average accuracy for referral items was 88.67%, followed by 73.74% for pathology values, 74.66% for radiology, 86.72% for clinical chemical laboratory, 75.97% for history, and 90.43% for patient characteristics values.

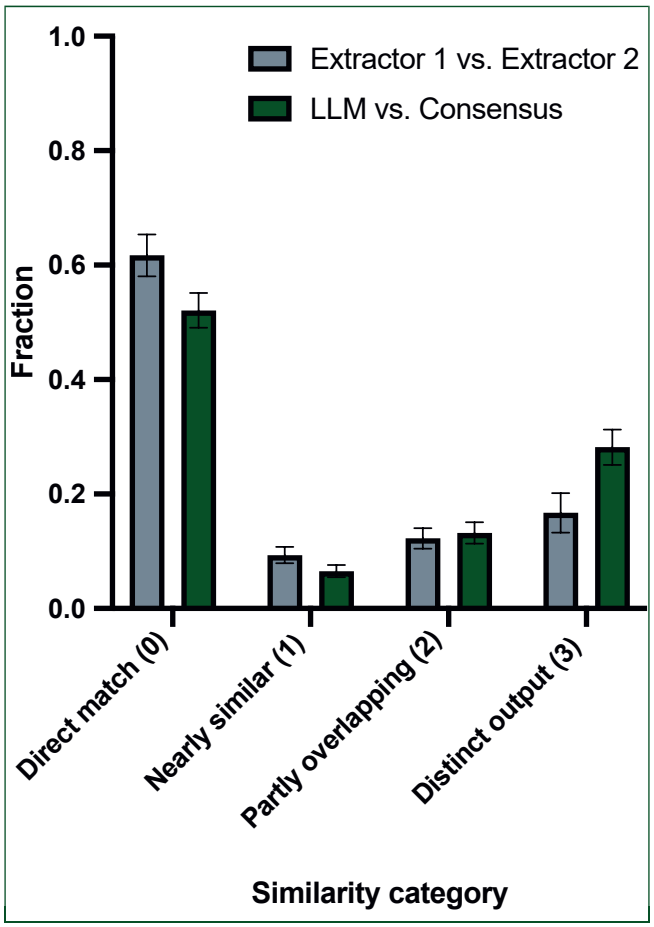


Figure 3. Distribution of initial mapping scores (mean with 95% confidence intervals calculated using the normal approximation) for two human extractors (blue) and the LLM extraction compared to the human consensus (green) before the multiclass evaluation rounds by experts. For each patient ($n = 60$), the fractions of extracted values falling into each mapping category (0-3) were first averaged at the patient level; means and confidence intervals were subsequently computed across patients. Mapping scores range from 0 to 3, where 0 indicates a direct match, 1 a minor disagreement (e.g. formatting differences), 2 a partial disagreement (e.g. incomplete information), and 3 a substantially different output without any overlap. A similar pattern is observed in initial agreement between the human extractors (blue) and the LLM compared to the consensus (green). LLM, large language model.

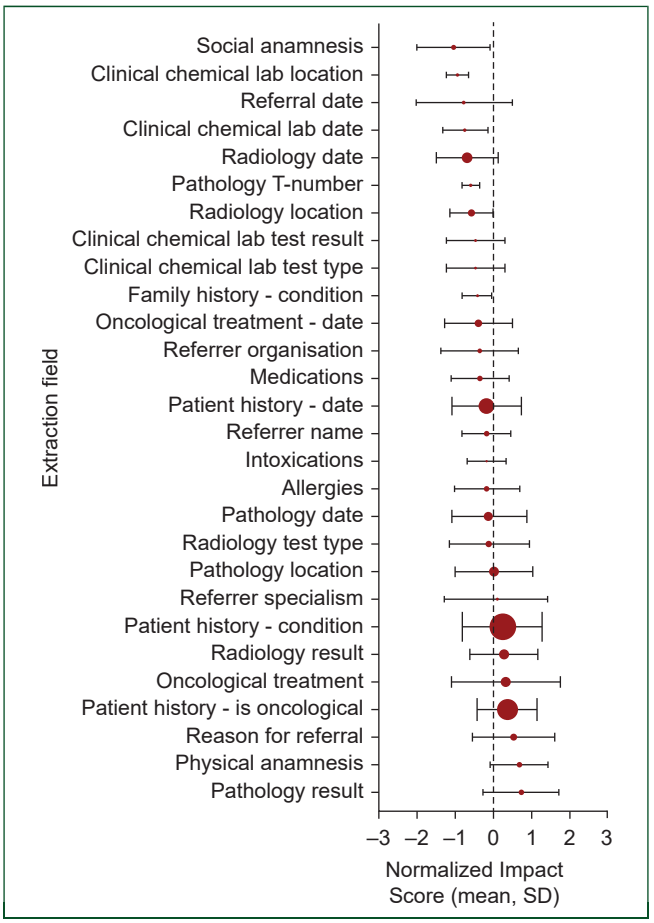


Figure 4. Normalized impact score per extraction category. The mean impact scores per category are shown, where the central dot size of each data point represents sample size, ranging from 2 to 70, and the error bars represent the standard deviation. The distribution is centered at 0, so the dotted line represents the average per-error impact. Fields to the right (left) hand side of this boundary can be viewed as having relatively high (low) impact errors.

Hallucinations were often misclassified

Initially, 65 model responses were classified as hallucinations. These were double-checked against the original documents and revised by evaluators. In the end, only four were confirmed as genuine hallucinations, all occurring within a single patient's pathology results. These involved nonexisting T-numbers used for identification of the pathology report (e.g. TCH22-5678) with atypical formatting (three letter identifiers instead of single letter identifiers, and sequential numbering), making them clearly detectable as incorrect or placeholder data. The impact scores for these four hallucinations were all 1, indicating low impact.

Time savings in the extraction step

Manual extraction of the values of the 29 categories took an average of 8.6 minutes [standard deviation (SD) 4.92] per patient. In contrast, LLM-powered extraction produced results in an average of 1.9 minutes (SD 1.11) per patient under identical input conditions (Supplementary Figure S4, available at https://doi.org/10.1016/j.esmorw.2026.100718). The latter time reflects only the extraction step by the model, not the full process of taking the LLM's results and using them in preparation for the first consultation. Nonetheless, this demonstrates that LLM-powered data extraction has the potential to speed up that data extraction task itself, alongside the capacity for continuous operation.

DISCUSSION

Summary of findings

This validation study assessed the accuracy and clinical impact of real-world clinical data extraction by an LLM within the HNC department of a specialized comprehensive cancer center. We found that the expert-assessed accuracy of the LLM data extraction ranged from 74% to 90% per category. We observed significant interobserver variability (29.0%) between the two human extractors, highlighting the inherent subjectivity in interpreting clinical documents. This natural variability suggests that, to be clinically useful, accuracy for data extraction should not be required to be

100%, as long as patient safety is ensured. Of the 2555 extracted and evaluated values, 751 were flagged as potential discrepancies with the human baseline standard. These 751 values were further evaluated and scored for their impact on the preparation of the initial consultation by expert evaluators. A total of 68 values were regarded as high-impact errors, with a large interevaluator variability (from 0 to 61). Hallucinations were detected in only 0.16% of extractions, all of which were regarded as low impact. In addition, the LLM reduced extraction time from 8.6 minutes (manual) to 1.9 minutes. This demonstrates the potential of LLM-assisted extraction to support clinical workflows, provided that human oversight is maintained.

Accuracy of LLM extraction

Van Veen et al.[14] showed that AI can outperform humans in clinical note summarization. Although extracted data forms the basis of our generated patient summaries, the concept of data extraction versus summarization is clearly distinct and therefore complex to compare. We expect that, due to the nature of the task of data extraction, with less reasoning compared to the generation of text, we observed fewer hallucinations. The observed accuracy is in line with that of Huang et al.,[15] who reported an overall accuracy of 89% in extracting pathological classifications in a lung cancer dataset.

The present study observed that the extraction accuracy varies across different extraction categories. The pathology and radiology categories showed lower point estimates and broader CIs, suggesting greater variability and less stable performance in these extraction domains. Pathology reports pose unique challenges due to their unique structure and the presence of multiple relevant dates (e.g. biopsy date, receipt date, procedure date, and report date), which can lead to confusion by both human and machine interpreters. In addition, these reports—in line with radiology reports—are often summarized in the referral letter, leading to duplicates or confusion regarding dates or locations during the extraction process.

Comparison with alternative AI approaches

In the past, various alternative NLP approaches have been used for clinical information extraction, including rule-based systems and classical machine-learning pipelines. Although these methods offer transparency and reproducibility, they require substantial manual configuration and typically generalize poorly to the variability of real-world clinical documentation.[16] LLMs address these limitations by offering greater flexibility, strong performance on previously unseen phrasing, and the ability to capture long-range semantic dependencies without task-specific feature engineering. At the same time, they introduce their own challenges: high computational and energy demands, sensitivity to prompt design, and reduced interpretability compared with rule-based or traditional statistical models (requiring explainability). They also require careful governance to ensure privacy, consistency, and alignment with clinical expectations.[17] Within oncology, platforms like MSK-CHORD (modular machine learning/NLP integrating structured and unstructured EHRs to support registry-grade datasets)[18] and mCODEGPT (zero-shot extraction aligned to mCODE)[19] demonstrate domain-aligned performance, and oncology-adapted language models (e.g. CancerLLM[20]) suggest benefits from task-specific adaptation. In contrast, our study deliberately uses a general-purpose model, showing that acceptable baseline performance can be achieved without oncology-specific fine-tuning. Given that domain adaptation can be slow and may lag rapid model advances, we emphasize, consistent with European Society for Medical Oncology, Guidance for Reporting Oncology real-World Evidence and ESMO Guidance on the Use of Large Language Models in Clinical Practice, that prospective clinical validation, oversight, and safety monitoring should remain the primary criteria for deployment, irrespective of the modeling strategy.[21]

Human variability

An unexpected outcome was the ambiguity between humans, for which we identified three distinct types. First, interobserver variability was observed during manual extraction. Despite ultimately achieving consensus, the process required extensive time and discussion, highlighting the inherent challenges of human agreement. Second, we observed inconsistencies in the classification of the extracted values into predefined categories. These differences were particularly pronounced in lengthy and complex documents, where heterogeneity in structure, formatting, and chronology substantially increased cognitive load and the risk of misinterpretation or omission of relevant content.

Third, the uncertainty of what constitutes relevant information led to discrepancies in the impact assessment of errors made by the model. Notably, one evaluator's judgments deviated significantly from those of the other five evaluators, which fits the real-world setting of this study. Prior literature shows that health care professionals often, consciously or unconsciously, rely on heuristics during clinical tasks.[22] It is plausible that the above influenced the generation of the human baseline and subsequent evaluations.

Strengths and limitations

A key strength of this study is the use of real-world clinical data and the involvement of practicing physicians in both extraction and evaluation. In addition, our findings demonstrate that, although LLMs are predominantly trained on vast amounts of English-language corpora,[23] they are also capable of processing Dutch clinical documents effectively. Nevertheless, several limitations should be acknowledged. The small number of human extractors limited our ability to statistically assess interobserver variability. Due to the presence of this variability, we should search for other methods to model true variation in the data and allow inherent ambiguity in the extraction task.

Furthermore, while efforts were made to blind evaluators, it is possible that physicians recognized their patients. However, we hypothesize that they would not know the exact content of the letters, thereby minimizing potential bias. Finally, a weakness of this study is the question of generalizability. For example, patients with HNC are referred relatively quickly to Antoni van Leeuwenhoek as a tertiary center. In different specialties, we could expect, for example, patients with a longer disease history or more complex disease presentation. Further research is needed to evaluate LLMs' performance in other cases and subsequent specialties.

Recommendations for future research

Aside from expanding validation across specialties and patient populations, further research should focus on improving accuracy while maintaining or enhancing safety, including risks such as increased chart clutter, automation biases, and model collapse.[24]

Improvement of accuracy might be reached by optimizing prompt and model combinations to further reduce false positives and false negatives or by using novel developments in AI, such as the use of Agentic AI.[25] In parallel, efforts should be made regarding data formatting and structuring with the use of standardized language[26] and visualization to enhance usability in real-time clinical settings.[27]

Importantly, this study highlights a broader unresolved question: What constitutes relevant information? The observed differences in human assessment indicate that impact is not an intrinsic property of an error type but is instead context- and role-dependent. In our data, we found no consistent pattern suggesting that specific categories systematically produce higher-impact errors. Larger sample sizes and broader evaluator participation may help to clarify how different clinicians weigh the importance of various fields. In addition, pre-calibration of evaluators could also reduce variability, although such processes require dedicated methodological work. Overall, we suggest that acceptable performance thresholds should account for both the inherent ambiguity of each category and the potential clinical consequences of errors.

Conclusion

This study demonstrates, in a clinically grounded validation design, the proof of concept that general-purpose LLMs can be effectively used for structured data extraction in clinical settings. Although the results are encouraging, improved reasoning and contextual handling could further strengthen performance. Continued validation in diverse clinical settings will be essential to ensure safe and consistent use in practice. Together, these findings suggest that LLMs can reduce administrative burden and facilitate secondary use of clinical data, contributing to the digital transformation of health care.

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DISCLOSURE

The authors declare the following potential conflict of interest: VS, NA, AR, and RK are employees of kaiko.ai, which may have a commercial interest in the outcomes of this research.

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